Dissertation / Project / Project Work Title: Evaluating Graph-Based Retrieval-Augmented Generation: Comparative Analysis and Quality Improvements over Standard RAG Systems

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1. Abstract

This project focuses on evaluating Graph-Based Retrieval-Augmented Generation (Graph-RAG) systems and comparing them with standard vector-based RAG approaches. The motivation is to improve answer accuracy and reasoning for complex and multi-hop queries by explicitly modeling relationships between entities using a knowledge graph. The work involves building two comparable QA pipelines: one using traditional vector retrieval and another leveraging a graph database: followed by systematic evaluation on a common dataset. The expected outcome is a clear understanding of scenarios where graph-based retrieval provides measurable benefits over standard RAG systems.

2. Broad Area of Work

The project lies at the intersection of:

- Conversational AI - Information Retrieval (IR)
- Knowledge Representation and Knowledge Graphs
- Large Language Models (LLMs) and Retrieval-Augmented Generation

3. Problem Statement and Background

Standard RAG systems primarily rely on vector similarity search, which often struggles with relational reasoning, entity linking, and multi-hop queries. In practical applications, minor retrieval errors can propagate into incorrect or misleading answers. This project investigates whether augmenting retrieval with graph-structured knowledge can improve contextual grounding and reasoning quality, especially for queries requiring understanding of relationships between multiple facts.

4. Objectives of the Project

- Design and implement a standard RAG-based QA system using vector retrieval.
- Design and implement a graph-based RAG system using a knowledge graph.
- Use a common dataset to benchmark both systems fairly.
- Evaluate performance on complex and multi-hop queries.
- Compare accuracy, relevance, and failure modes of both approaches.

5. Dataset Description

The dataset selected for evaluation is a publicly available synthetic dataset of articles:

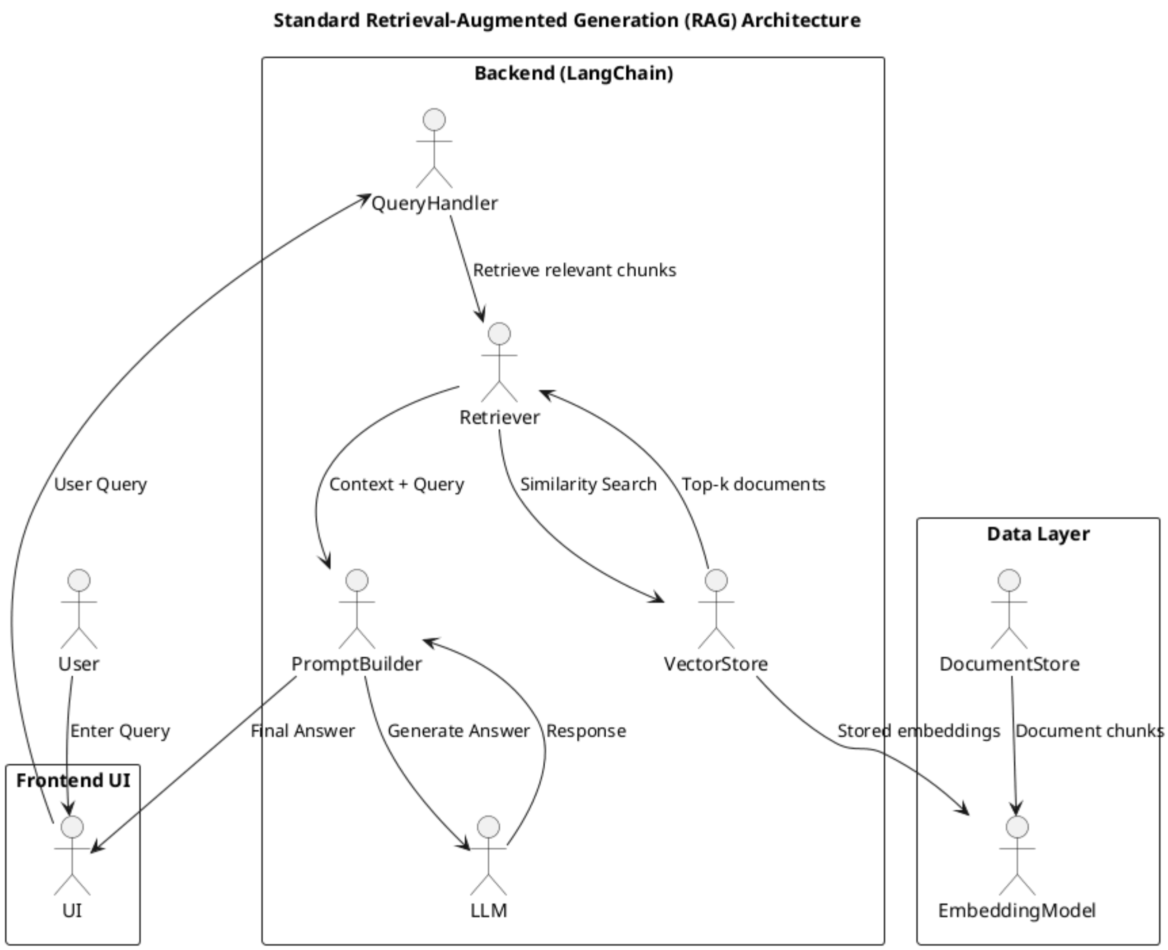
- Dataset: synthetic_articles.csv
- Source: https://github.com/dcarpintero/ai-engineering/blob/main/dataset/synthetic_articles.csv

The dataset contains structured synthetic articles suitable for controlled experimentation, allowing
consistent comparison between vector-based and graph-based retrieval pipelines.

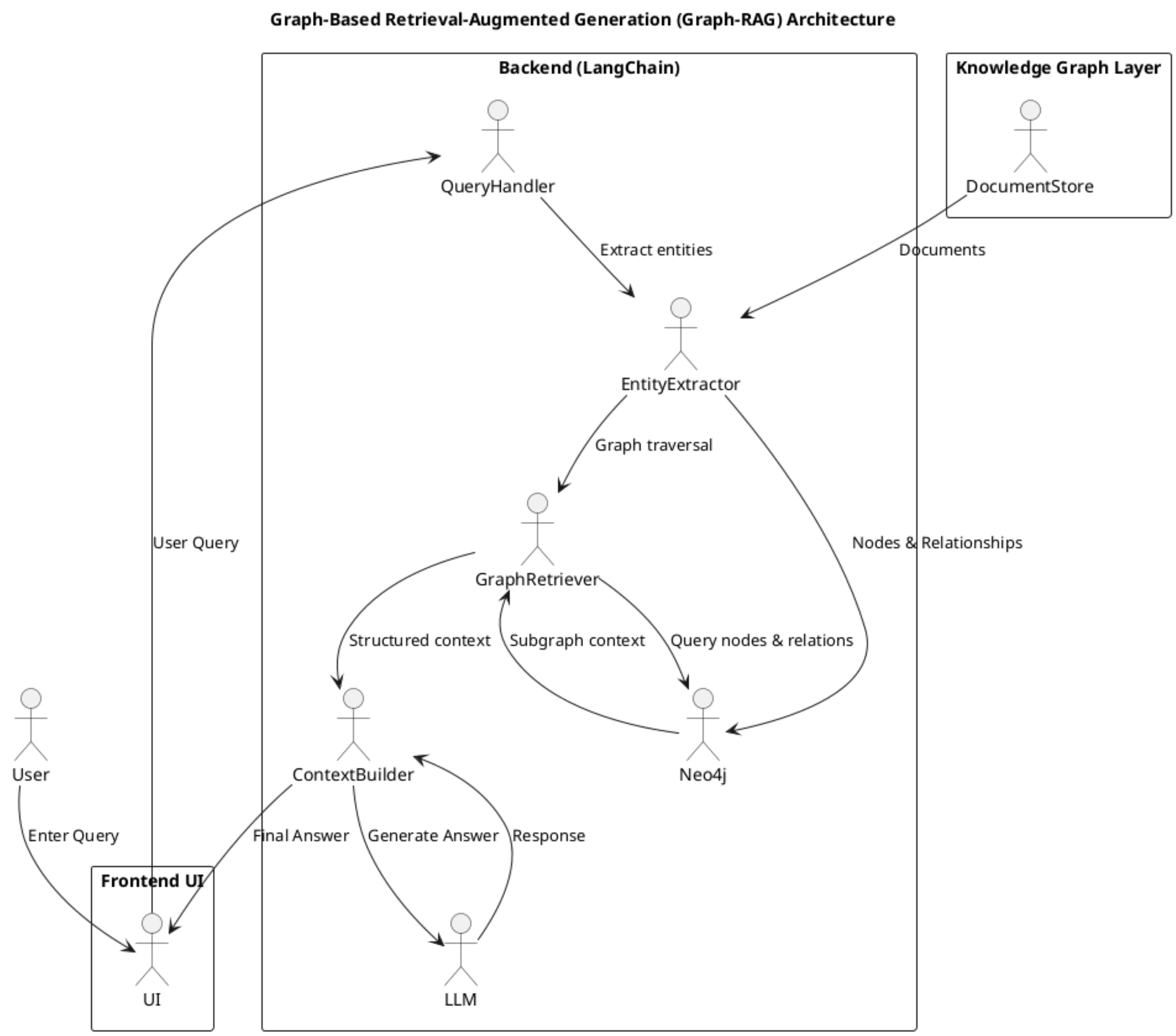
6. System Architecture Overview

The project consists of two parallel pipelines built on the same dataset and language model:

- Standard RAG Pipeline
  - Document ingestion and chunking
  - Embedding generation using pretrained embedding models
  - Vector storage and similarity-based retrieval
  - Answer generation using a pretrained LLM



- Graph-Based RAG Pipeline
  - Entity and relationship extraction from documents
  - Knowledge graph construction and storage
  - Graph-based retrieval using entity and relationship traversal
  - Context assembly and answer generation using a pretrained LLM



7. Tools and Technologies Used

- **Frameworks:** LangChain
- **Graph Database:** Neo4j
- **Vector Store:** Chroma
- **LLMs:** Pretrained open-source models (0.5B–1B range)
- **Embeddings:** Pretrained embedding models
- **Frontend:** Simple Streamlit UI for query input and answer display
- **Programming Language:** Python

8. Feasibility Study

The project is feasible within the given timeline based on the following factors:

- **Dataset Availability:** The selected synthetic dataset is publicly available, lightweight, and well-suited for rapid experimentation and benchmarking.
- **Tool Maturity:** LangChain and Neo4j are stable, well-documented tools that significantly reduce development overhead for RAG and graph-based systems.
- **Model Choice:** Use of pretrained LLMs and embedding models avoids the need for expensive training or fine-tuning, keeping computational requirements manageable.
- **System Scope:** The project focuses on prototype-level systems rather than production-scale deployment, keeping complexity controlled.
- **Frontend Simplicity:** A streamlit frontend is sufficient for evaluation and demonstration purposes.
- **Compute Constraints:** The systems are designed to run on local or modest compute resources, aligning with academic timelines.

9. Work Completed So Far

- Literature review on RAG and Graph-RAG approaches
- Finalization of problem statement and objectives
- Dataset selection and initial exploration
- Tool and framework finalization
- High-level system design for both pipelines

10. Work Planned for Remaining Period

- Implementation of standard RAG pipeline
- Implementation of graph-based RAG pipeline using Neo4j
- Query set preparation for evaluation
- Comparative evaluation and result analysis
- Report writing and final submission

11. Plan of Work (Revised Status as of Mid-Sem)

The following plan of work was proposed during the abstract phase. As of the mid-semester stage, only initial setup activities have been completed. The remaining phases are pending and will be executed strictly within the institute-defined timelines.

Phases	Start Date – End Date	Work to be done	Current Status
Literature review and outline	09 Nov 2025 – 24 Nov 2025	Read work on RAG and graph-based RAG, finalize problem statement, freeze tools and dataset, prepare outline.	**Completed**
Initial setup & dataset selection	25 Nov 2025 – 01 Dec 2025	Codebase setup, environment configuration, frontend scaffold, dataset selection and validation.	**Completed**
Standard RAG system development	02 Dec 2025 – 21 Dec 2025	Design and development of vector-based RAG pipeline including embeddings, retriever, and QA flow.	**Ongoing**
Graph-based RAG system development	02 Dec 2025 – 21 Dec 2025	Entity extraction, knowledge graph construction using Neo4j, and graph-based retrieval pipeline.	**Ongoing**
Testing	22 Dec 2025 – 11 Jan 2026	Software testing, query evaluation, user evaluation, and interim conclusions.	**Pending**
Evaluation and comparison	12 Jan 2026 – 24 Jan 2026	Comparative analysis of standard RAG and graph-based RAG, submit results for supervisor and examiner review.	**Pending**
Report writing and finalization	25 Jan 2026 – 01 Feb 2026	Final review and submission of dissertation.	**Pending**

12. Expected Outcomes

- A working comparison between standard RAG and graph-based RAG systems
- Quantitative and qualitative insights into retrieval and reasoning differences
- Clear identification of scenarios where graph-based retrieval offers advantages

13. Signatures

Signature of the Student:

Name: Shubhra Gadhwala
Date: 21/12/2025

Signature of the Supervisor:

Name: Darshan Derasari
Date: 21/12/2025